

Harish Kumar S

Associate Software Developer | AI/ML · MLOps · Data Platforms

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SUMMARY

Associate Software Developer at FactEntry (A SIX Company) building production AI and data platforms. Experienced in FastAPI microservices, Redis-based distributed processing, Kubernetes (K3s) with GitOps delivery, RAG systems, and computer vision, and now cloud data engineering on Databricks (data lake, data warehouse, MLflow). Integrated M.Tech in Data Science from VIT (2025).

EXPERIENCE

Associate Software Developer

Dec 2025 – Present

FactEntry Data Solutions (A SIX Company)

- Build and run the Data Mining Engine (DME), a production bond-extraction platform for corporate and municipal bonds on K3s (80+ CPU cores) with a human-in-the-loop validation UI.
- Designed Redis Streams job dispatch across multi-node workers, plus FastAPI microservices with live WebSocket job telemetry, role-based CPU tiering, and domain-restricted auth.
- Implemented GitOps delivery: GitLab CI image builds, Helm chart promotion, and Argo CD sync to staging.
- Working with SIX (parent company) to migrate DME to Databricks: data lake and data warehouse design, with MLflow for experiment and model tracking.

Associate Software Engineer Trainee Intern

Jun 2025 – Oct 2025

MIMASOFT Technologies

- Fine-tuned two YOLOv11 detection models in Roboflow for dental imaging: tooth detection (mAP@50 87.7%, precision 89.3%) and 14-class panoramic X-ray (mAP@50 70.3%).
- Built an inference CLI for Roboflow serverless (inference-sdk) and local model backends; contributed to backend API integration.

MLOps and Data Science Intern

Nov 2024 – May 2025

FactEntry Data Solutions (A SIX Company)

- Published language-processing-tool on PyPI (v0.2.7) for PDF language detection, OCR, and layout-based document clustering; delivered an advanced internal version (VGG16/ResNet clustering, LLaMA Vision) to FactEntry.
- Built Fixed Income Expert, a RAG chatbot over bond documents (SBERT, FAISS, Groq LLaMA 3, FastAPI + Streamlit), deployed as systemd services; contributed to DME prototype R&D.

Data Science Intern (ML and SDE)

Nov 2023 – Jan 2024

JP INFOTECH

- Built an extractive summarization and information-retrieval system using NLP-driven web scraping with English-to-SQL query translation (SQLite/MySQL), reaching 95% extraction accuracy.
- Built an applicant tracking tool that OCRs resumes (PyTesseract, OpenCV), matches them to job descriptions, and suggests missing keywords.

Applied Data Science Extern

May 2023 – Jul 2023

Smart Bridge - SmartInternz (Google Developers)

- Built a Random Forest + SVM ensemble for disease prediction on clinical-trial data, fed by a MongoDB Atlas (PyMongo) data pipeline.

PROJECTS

PAI PDF AI — Document Intelligence Platform | FastAPI, React, Elasticsearch, Ollama, PyMuPDF

Privacy-first analysis of legal and financial PDFs: local-LLM extraction of clauses, entities, and risks, with answers cited to exact source chunks from a vector knowledge base.

Recruiting Agent — AI-Powered ATS | FastAPI, React, SQLite, ChromaDB, Ollama, Groq

Nine-step LLM pipeline that OCRs resumes, matches them to job descriptions in ChromaDB, verifies claims, scores fit and authenticity, and generates evidence-based HR questions.

Trip Guide — Conversational Trip Planner | FastAPI, Google ADK, Gemini, Groq, Ollama, Leaflet

Multi-agent travel assistant: Google ADK orchestrator with weather, places, and stays specialists, SSE streaming chat, live trip board and OSRM route map, and Gemini to Groq to Ollama failover.

SKILLS

Languages & Backend: Python, SQL, C/C++, FastAPI, Flask, WebSockets, React, Streamlit

AI / ML & GenAI: PyTorch, TensorFlow, scikit-learn, Deep Learning, NLP, Computer Vision, RAG, LLM Agents (Google ADK), LangChain, FAISS, ChromaDB, Sentence Transformers, Ollama (fine-tuning), Groq, YOLOv11, Roboflow, OpenCV, Tesseract OCR

Data & Cloud: Databricks, Data Lake, Data Warehouse, ETL Pipelines, Microsoft Azure, GCP, PostgreSQL, Redis Streams, MongoDB, Elasticsearch

MLOps & DevOps: Kubernetes (K3s), Docker / Compose, Helm, Argo CD, GitLab CI, GitHub Actions, GitOps, MLflow, Linux / systemd

EDUCATION

Integrated M.Tech, Computer Science and Engineering (Data Science)

2020 – 2025

Vellore Institute of Technology | CGPA 7.8

Class XII: 79%, THIRU G V C Higher Secondary School (2020) | Class X: 87.8%, St Joseph Matriculation Higher Secondary School (2018)

CERTIFICATIONS

Microsoft Azure AI Fundamentals • Applied Data Science (SmartInternz + Google) • Power BI & Excel (Coursera, 4 courses)